

Xigatin®

Film Coated Tablet 100 mg

Active Ingredient:

Each film coated tablet contains:

Sitagliptin Phosphate Monohydrate 128.5 mg
(eq. to 100 mg Sitagliptin)

Pharmacotherapeutic group:

Drugs used in diabetes, Dipeptidyl peptidase 4 (DPP-4) inhibitors, ATC code: A10BH01.

Pharmacodynamics:

Mechanism of action

Sitagliptin is a member of a class of oral anti-hyperglycaemic agents called dipeptidyl peptidase 4 (DPP-4) inhibitors. The improvement in glycaemic control observed with this medicinal product may be mediated by enhancing the levels of active incretin hormones. Incretin hormones, including glucagon-like peptide-1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP), are released by the intestine throughout the day, and levels are increased in response to a meal. When blood glucose concentrations are normal or elevated, GLP-1 and GIP increase insulin synthesis and release from pancreatic beta cells by intracellular signaling pathways involving cyclic AMP. With higher insulin levels, tissue glucose uptake is enhanced. In addition, GLP-1 lowers glucagon secretion from pancreatic alpha cells. Decreased glucagon concentrations, along with higher insulin levels, lead to reduced hepatic glucose production, resulting in a decrease in blood glucose levels.

For both GLP-1 and GIP, stimulation of insulin release is enhanced as glucose rises above normal concentrations. The activity of GLP-1 and GIP is limited by the DPP-4 enzyme, which rapidly hydrolyzes the incretin hormones to produce inactive products. Sitagliptin prevents the hydrolysis of incretin hormones by DPP-4, thereby increasing plasma concentrations of the active forms of GLP-1 and GIP. By enhancing active incretin levels, sitagliptin increases insulin release and decreases glucagon levels in a glucose-dependent manner. In patients with type 2 diabetes with hyperglycaemia, these changes in insulin and glucagon levels lead to lower haemoglobin A_{1c} (HbA_{1c}) and lower fasting and postprandial glucose concentrations. The glucose-dependent mechanism of sitagliptin is distinct from the mechanism of sulphonylureas, which increase insulin secretion even when glucose levels are low and can lead to hypoglycaemia in patients with type 2 diabetes and in normal subjects. Sitagliptin is a potent and highly selective inhibitor of the enzyme DPP-4 and does not inhibit the closely-related enzymes DPP-8 or DPP-9 at therapeutic concentrations.

Pharmacokinetics:

Absorption

Following oral administration of a 100-mg dose, sitagliptin was rapidly absorbed, with peak plasma concentrations (median T_{max}) occurring 1 to 4 hours post-dose, mean plasma AUC of sitagliptin was 8.52 µM·hr, C_{max} was 950 nM. The absolute bioavailability of sitagliptin is approximately 87 %. Since co-administration of a high-fat meal with sitagliptin had no effect on the pharmacokinetics, Xigatin may be administered with or without food.

Plasma AUC of sitagliptin increased in a dose-proportional manner. Dose-proportionality was not established for C_{max} and C_{24hr} (C_{max} increased in a greater than dose-proportional manner and C_{24hr} increased in a less than dose-proportional manner).

Distribution

The mean volume of distribution at steady state following a single 100-mg intravenous dose of sitagliptin is approximately 198 litres. The fraction of sitagliptin reversibly bound to plasma proteins is low (38 %).

Biotransformation

Sitagliptin is primarily eliminated unchanged in urine, and metabolism is a minor pathway. Approximately 79 % of sitagliptin is excreted unchanged in the urine. Following a [¹⁴C]sitagliptin oral dose, approximately 16 % of the radioactivity was excreted as metabolites of sitagliptin.

Elimination

Following administration of an oral [¹⁴C]sitagliptin dose, approximately 100 % of the administered radioactivity was eliminated in faeces (13 %) or urine (87 %) within one week of dosing. The apparent terminal t_{1/2} following a 100-mg oral dose of sitagliptin was approximately 12.4 hours. Sitagliptin accumulates only minimally with multiple doses. The renal clearance was approximately 350 mL/min. Elimination of sitagliptin occurs primarily via renal excretion and involves active tubular secretion.

Characteristics in patients

Hepatic impairment

No dose adjustment for Xigatin® is necessary for patients with mild or moderate hepatic impairment (Child-Pugh score ≤ 9). However, because sitagliptin is primarily renally eliminated, severe hepatic impairment is not expected to affect the pharmacokinetics of sitagliptin.

Elderly

No dose adjustment is required based on age.

Paediatric population

No studies with sitagliptin have been performed in paediatric patients with age <10 years.

Other patient characteristics

No dose adjustment is necessary based on gender, race, or body mass index (BMI).

Indication:

Monotherapy

Xigatin® is indicated as an adjunct to diet and exercise to improve glycemic control in patients with type 2 diabetes mellitus.

Combination with metformin

Xigatin® is indicated in patients with type 2 diabetes mellitus to improve glycemic control in

combination with metformin as initial therapy or when the single agent alone, with diet and exercise, does not provide adequate glycemic control. Initial combination therapy or maintenance of combination therapy may not be appropriate for all patients. These management options are left to the discretion of the health care provider.

Combination with a sulphonylurea

Xigatin® is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with a sulphonylurea when treatment with maximal tolerated dose of sulphonylurea alone, with diet and exercise, does not provide adequate glycemic control and when metformin is inappropriate due to contraindications or intolerance.

Combination with metformin and a sulphonylurea

Xigatin® is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with metformin and a sulphonylurea when dual therapy with these two agents and with diet and exercise does not provide adequate glycemic control.

Combination with a peroxisome proliferator-activated receptor gamma (PPAR γ) agonist

Xigatin® is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with a PPAR γ agonist (i.e. thiazolidinediones) when diet and exercise, plus the single agent do not provide adequate glycemic control.

Combination with metformin and a PPAR γ agonist

Xigatin® is indicated in patients with type 2 diabetes mellitus to improve glycemic control in combination with metformin and a PPAR γ agonist (i.e., thiazolidinediones) when dual therapy with these agents, with diet and exercise, does not provide adequate glycemic control.

Combination with Insulin

Xigatin® is indicated in patients with type 2 diabetes mellitus as an adjunct to diet and exercise to improve glycemic control in combination with insulin (with or without metformin).

Recommended Dose:

Posology (adults)

The recommended dose of Xigatin® is 100 mg once daily as monotherapy or as combination therapy with metformin, a sulphonylurea, insulin (with or without metformin), a PPAR γ agonist (i.e., thiazolidinediones), metformin plus a sulphonylurea, or metformin plus a PPAR γ agonist. Xigatin® can be taken with or without food.

When Xigatin® is used in combination with a sulphonylurea or with insulin, a lower dose of sulphonylurea or insulin may be considered to reduce the risk of sulphonylurea- or insulin-induced hypoglycaemia. (See PRECAUTIONS, Hypoglycemia in Combination with a Sulphonylurea or with Insulin)

Patients with Renal Impairment

Because there is a dosage adjustment based upon renal function, assessment of renal function is recommended prior to initiation of Xigatin® and periodically thereafter.

For patients with mild renal impairment (estimated glomerular filtration rate [eGFR] ≥ 60 mL/min/1.73 m² to < 90 mL/min/1.73 m²), no dosage adjustment for Xigatin® is required.

For patients with moderate renal impairment (eGFR ≥ 45 mL/min/1.73 m² to < 60 mL/min/1.73 m²), no dosage adjustment for Xigatin® is required.

For patients with moderate renal impairment (eGFR ≥ 30 mL/min/1.73 m² to < 45 mL/min/1.73 m²), the dose of Xigatin® is 50 mg once daily.

For patients with severe renal impairment (eGFR ≥ 15 mL/min/1.73 m² to < 30 mL/min/1.73 m²) or with end-stage renal disease (ESRD) (eGFR < 15 mL/min/1.73 m²), including those requiring hemodialysis or peritoneal dialysis, the dose of Xigatin® is 25 mg once daily. Xigatin® may be administered without regard to the timing of dialysis.

Pediatric population

Xigatin® should not be used in children and adolescents 10 to 17 years of age because of insufficient efficacy. Xigatin® has not been studied in pediatric patients under 10 years of age.

Route of Administration

Oral administration.

The tablet should be swallowed whole with sufficient liquid and can be taken with or without food.

Contraindication:

- Hypersensitivity to the active ingredient or any of the excipients.
- Pregnancy and lactation.

Warnings and Precautions:

General

Xigatin® should not be used in patients with type 1 diabetes or for the treatment of diabetic ketoacidosis.

Acute pancreatitis

Use of DPP-4 inhibitors has been associated with a risk of developing acute pancreatitis. Patients should be informed of the characteristic symptom of acute pancreatitis: persistent, severe abdominal pain. Resolution of pancreatitis has been observed after discontinuation of sitagliptin (with or without supportive treatment), but very rare cases of necrotising or haemorrhagic pancreatitis and/or death have been reported. If pancreatitis is suspected, Xigatin® and other potentially suspect medicinal products should be discontinued; if acute pancreatitis is confirmed, Xigatin® should not be restarted. Caution should be exercised in patients with a history of pancreatitis.

Hypoglycaemia when used in combination with other anti-hyperglycaemic medicinal products

Hypoglycaemia has been observed when sitagliptin was used in combination with insulin or a sulphonylurea. Therefore, to reduce the risk of hypoglycaemia, a lower dose of sulphonylurea or insulin may be considered.

Renal impairment

Sitagliptin is renally excreted. To achieve plasma concentrations of sitagliptin similar to those in patients with normal renal function, lower dosages are recommended in patients with GFR < 45 mL/min, as well as in ESRD patients requiring haemodialysis or peritoneal dialysis. When considering the use of sitagliptin in combination with another anti-diabetic medicinal product, its conditions for use in patients with renal impairment should be checked.

Hypersensitivity reactions

Serious hypersensitivity reactions in patients treated with sitagliptin have been reported. These reactions include anaphylaxis, angioedema, and exfoliative skin conditions including Stevens-Johnson syndrome. Onset of these reactions occurred within the first 3 months after initiation of treatment, with some reports occurring after the first dose. If a hypersensitivity reaction is suspected, Xigatin® should be discontinued. Other potential causes for the event should be assessed, and alternative treatment for diabetes initiated.

Bullous pemphigoid

There have reports of bullous pemphigoid in patients taking DPP-4 inhibitors including sitagliptin. If bullous pemphigoid is suspected, Xigatin® should be discontinued.

Sodium

This medicinal product contains less than 1 mmol sodium (23 mg) per tablet, that is to say essentially 'sodium-free'.

Effect on ability to drive or operate machinery:

Sitagliptin has no or negligible influence on the ability to drive and use machines. However, when driving or using machines, it should be taken into account that dizziness and somnolence have been reported.

In addition, patients should be alerted to the risk of hypoglycaemia when sitagliptin is used in combination with a sulphonylurea or with insulin.

Interactions with Other Medicaments:

Effects of other medicinal products on sitagliptin

Metformin: Co-administration of multiple twice-daily doses of 1,000 mg metformin with 50 mg sitagliptin did not meaningfully alter the pharmacokinetics of sitagliptin in patients with type 2 diabetes.

Ciclosporin: Co-administration of a single 100 mg oral dose of sitagliptin and a single 600 mg oral dose of ciclosporin increased the AUC and C_{max} of sitagliptin by approximately 29 % and 68 %, respectively. These changes in sitagliptin pharmacokinetics were not considered to be clinically meaningful. The renal clearance of sitagliptin was not meaningfully altered. Therefore, meaningful interactions would not be expected with other p-glycoprotein inhibitors.

Effects of sitagliptin on other medicinal products

Digoxin: Sitagliptin had a small effect on plasma digoxin concentrations. Following administration of 0.25 mg digoxin concomitantly with 100 mg of sitagliptin daily for 10 days, the plasma AUC of digoxin was increased on average by 11 %, and the plasma C_{max} on average by 18 %. No dose adjustment of digoxin is recommended. However, patients at risk of digoxin toxicity should be monitored for this when sitagliptin and digoxin are administered concomitantly.

Pregnancy and lactation

Pregnancy

There are no adequate data from the use of sitagliptin in pregnant women. The potential risk for humans is unknown. Due to lack of human data, Xigatin® should not be used during pregnancy.

Breast-feeding

It is unknown whether sitagliptin is excreted in human breast milk. Animal studies have shown excretion of sitagliptin in breast milk. Xigatin® should not be used during breast-feeding.

Adverse/Undesirable Effects

Tabulated list of adverse reactions

Adverse reactions are listed below (Table 1) by system organ class and frequency. Frequencies are defined as: very common (≥ 1/10); common (≥ 1/100 to < 1/10); uncommon (≥ 1/1,000 to < 1/100); rare (≥ 1/10,000 to < 1/1,000); very rare (< 1/10,000) and not known (cannot be estimated from the available data).

Adverse reaction	Frequency of adverse reaction
Blood and lymphatic system disorders	
thrombocytopenia	Rare
Immune system disorders	
hypersensitivity reactions including anaphylactic responses	Frequency not known
Metabolism and nutrition disorders	
hypoglycaemia	Common
Nervous system disorders	
headache	Common
dizziness	Uncommon
Respiratory, thoracic and mediastinal disorders	
interstitial lung disease	Frequency not known
Gastrointestinal disorders	
constipation	Uncommon

vomiting	Frequency not known
acute pancreatitis	Frequency not known
fatal and non-fatal haemorrhagic and necrotizing pancreatitis	Frequency not known
<i>Skin and subcutaneous tissue disorders</i>	
pruritus	Uncommon
angioedema	Frequency not known
rash	Frequency not known
urticaria	Frequency not known
cutaneous vasculitis	Frequency not known
exfoliative skin conditions including Stevens-Johnson syndrome	Frequency not known
bullous pemphigoid	Frequency not known
<i>Musculoskeletal and connective tissue disorders</i>	
arthralgia	Frequency not known
myalgia	Frequency not known
back pain	Frequency not known
arthropathy	Frequency not known
<i>Renal and urinary disorders</i>	
impaired renal function	Frequency not known
acute renal failure	Frequency not known

Table 1: Side/ Adverse Effects of Xigatin®

Overdose and Treatment

In the event of an overdose, it is reasonable to employ the usual supportive measures, e.g., remove unabsorbed material from the gastrointestinal tract, employ clinical monitoring (including obtaining an electrocardiogram), and institute supportive therapy if required. Sitagliptin is modestly dialysable. Approximately 13.5 % of the dose was removed over a 3- to 4-hour haemodialysis session. Prolonged haemodialysis may be considered if clinically appropriate. It is not known if sitagliptin is dialysable by peritoneal dialysis.

Storage Conditions

Store at temperature below 30°C. Protect it from light and moisture.

Dosage forms and packaging available

100mg film coated tablet, 10's x 3

Shelf Life:

3 years

Product Description:

An orange brown film coated round tablet, one side impressed with '100'.



Manufacturer and Product Registration Holder:

Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)

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